A Comparative Case Study of EMG-Driven Controllers in Transtibial Prostheses

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***Abstract*— Lower limb amputation greatly affects quality of life by restricting functional mobility. Despite advancements in prosthetic design, powered transtibial prostheses still have limitations in user control and adaptability to dynamic environments. This research presents a comparative analysis between a novel electromyography (EMG)-driven variable impedance controller (VIC) and a hybrid controller (HC) that integrates a volitional EMG-driven musculoskeletal model with a finite-state machine impedance controller. A Hill-type muscle model was used to model the gastrocnemius and tibialis anterior muscles. Biomechanical testing was conducted with a transtibial amputee to assess the controllers' performance across various tasks, including ambulation on level ground, stairs, and ramps, using EMG signals from the residual limb. Results demonstrated that the VIC provided more repeatable performance, perceived control, and power output. Notable effect sizes for peak power, observed in ramp ascent (Cohen's d = −1.04) and high-speed level ground walking (Cohen's d = −2.92), illustrate robust differences in joint-level output even when walking speeds and cadences were comparable. The greater predictability of the VIC led the user to feel more in control and comfortable throughout the various activities. On the other hand, the HC controller performed better in enabling more seamless transitions between gait subphases, particularly during stair ascent, which led to a significantly higher ROM (18.63 ± 1.53 deg vs. 12.43 ± 1.86 deg) and nearly double peak power compared to the VIC. This comparison lays the groundwork for future research into optimizing EMG-based control strategies that adapt to both biomechanical demands and user preferences.**

***Index Terms*— Electromyography, neuromuscular model, variable impedance controller, prosthetic control.**

Received 20 May 2025; revised 21 July 2025; accepted 20 August 2025. Date of publication 25 August 2025; date of current version 2 September 2025. This work was supported by the Technology Development Fund at the Icelandic Center for Research under Grant 2215370-0611. *(Corresponding author: M. Abdelbar.)*

This work involved human subjects in its research. Approval of all ethical and experimental procedures and protocols was granted by Icelandic National Bioethics Committee under Application No. CII2019061252.

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Digital Object Identifier 10.1109/TNSRE.2025.3602296

I. Introduction

LOWER limb amputation has an impact on people's quality of life since it limits their ability to perform functional tasks [1], [2]. These individuals are commonly prescribed passive prostheses, which are incapable of restoring enough of the physiological function of the joint [3]. To address this challenge, powered prostheses have been developed to produce torque and power that are comparable to those of a non-disabled joint [4], [5], [6]. While the mechatronics of these powered robotic prostheses has advanced in terms of intelligence, practicality, and accessibility [7], [8], they still face limitations. These devices often exhibit restricted functionality due to their pre-programmed, autonomous nature, as they are typically designed for specific tasks and activities. Although control algorithms have become increasingly complex in efforts to improve adaptability, direct user input and real-time volitional control remain limited [5], [9], [10], [11], [12]. Machine learning algorithms can enable greater variability and shaping of the controller by not entirely depending on heuristically determined state-switching criteria. However, machine learning controllers still face challenges in allowing users to navigate through crowds effectively, avoid obstacles, and adapt to time-varying stepping patterns [13], [14], [15], [16].

Biomedical researchers have been driven to explore ways to enable users to control prosthetic limbs as intuitively as their biological ones [17], [18], [19], [20], [21]. The main idea involves establishing a direct connection between the prosthesis actuators and the human nervous system, enabling users to transmit commands to the prosthetic limb. Electromyography (EMG) offers a promising approach by decoding peripheral efferent signals from the muscles of the residual limb [22]. In recent years, EMG control has gained popularity as a method to achieve voluntary control of lower limb prostheses [22], [23], [24], [25], [26]. Its feasibility stems from the human motor control system's remarkable flexibility and adaptability to varying tasks and environments.

Implementation of EMG control is commonly achieved via either supervisory or direct EMG control. Supervisory control is a type of discrete control in which a pattern classifier identifies the user's intended locomotion mode and selects the appropriate control strategy [20], [27], [28]. This approach requires a substantial amount of user-specific training

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data and is prone to classification errors, which can limit reliability [22]. On the other hand, direct EMG control continuously correlates the signal from the muscle with the force generated [29]. By directly mapping EMG input to control output and mimicking biological musculoskeletal (MSK) systems, this method supports reactive and non-cyclic activities. Furthermore, continuous volitional control of robotic ankle prostheses may be achieved using an EMG-driven MSK model that is activated by the relative EMG signal magnitude of residual agonist-antagonist muscle pairs [30], [31].

However, maintaining continuous control can increase the cognitive and physical load on the user, potentially leading to fatigue, distraction, or misinterpretation of intent, resulting in decreased performance and safety [32], [33]. Acknowledging these concerns, recent developments aim to use the enhanced user control provided by volitional control and the consistency and repeatability of non-volitional control. For example, Wang et al. [34] developed a controller that applied proportional EMG control only during powered plantar flexion for direct volitional modulation of the mid-to-late phase of stance. The presented case was, however, limited to level-ground walking. The controller developed by Kannape and Herr [35] enabled transtibial amputees to modify the push-off of their prosthesis proportionally. While it improved stair ambulation and transition from level walking to stairs, this controller's modulation was limited to plantar flexion and lacked advanced biomechanical testing or ramp evaluation. More recently, Posh et al. [36] explored the integration of finite-state machines (FSM) with direct myoelectric control throughout the gait cycle for a transtibial prosthesis. However, the hybrid volitional controller functioned more like either FSM impedance control or direct myoelectric control, rather than combining both features. Additionally, the controller was not assessed on stairs and ramps.

Another approach to integrate EMG signals into the FSM is to employ a variable impedance controller (VIC) to modulate the stiffness value of the impedance control law for FSM subphases, based on the relative co-activity from residual muscles [23], [37] . Physiological models, such as Hill-based models, which include both static and dynamic features, are commonly chosen to estimate joint stiffness and torque at the same time [38], [39]. These models are generally accurate and subject-specific [40]; however, their complexity poses challenges for real-time implementation [38].

Recent examples include a model using functional electrical stimulation (FES) to modulate ankle torque and stiffness simultaneously [38]. The model consists of four subparts (a joint stiffness model, an MSK geometry model, a Hill-based muscle-tendon model, and an FES-to-activation model) and showed high accuracy in estimating torque and stiffness for the gastrocnemius (GAS) and tibialis anterior (TA) muscles. Important study limitations were reflected in the difference between estimated and measured timing of torques and stiffness, likely due to electromechanical delay and the complexity of MSK models. Another similar approach used EMG instead of FES [41], allowing estimation of ankle torque and stiffness under both perturbed and unperturbed conditions.

Non-physiological models are simpler but only address the muscle's input-output mapping relationship, making it a challenge to evaluate joint stiffness [38]. For example, using a hyperbolic tangent function can model the nonlinear relationship between joint stiffness and muscle activity [42], [43], but does not account for joint angle, which is a key factor influencing joint responses during movement. In contrast, MSK models include the joint angle as an input to provide a more accurate estimation of stiffness [38], [41].

In our previous work [44], we introduced a hybrid controller (HC). During the stance phase, it utilized the FSM impedance controller combined with an EMG-driven neuromuscular model (NM), while during the swing phase, it functioned as a position controller. A study with a transtibial amputee across various tasks showed that the participant felt more engaged and was more able to control motion through muscle activation when using the HC as compared to FSM control without EMG. However, the study identified some limitations, including excessive plantar flexion during the controlled plantar flexion (CP) subphase indicated by ankle angle measurements and a lack of comprehensive biomechanical data analysis. Our subsequent study [45], mitigated excessive plantar flexion by disabling EMG control during the CP subphase, which resulted in a more natural gait pattern. While these adjustments were promising, the single-subject study was limited to testing during stair ascent and level ground walking, and notable limitations remained. Implementing the full NM model in real time proved technically challenging, primarily due to the mathematical complexity of the equations involved, which placed significant demands on processing resources and real-time computation. Additionally, because of this complexity, prosthesis behavior became more variable, making the system less predictable for users and sensitive to parameter changes. Additionally, simply summing the FSM impedance torque and the NM torque often led to excessively high-power outputs, which did not always yield intuitive or controllable joint responses; sometimes, the participant found it difficult to anticipate the combined output during dynamic activities.

Therefore, the purpose of this study was to address the limitations of our previous hybrid controller [44], [45], such as real-time computational challenges, excessive power generation, and lack of repeatability, by a) developing a novel EMG-driven controller that modulates stiffness for FSM subphases during the stance phase, b) comprehensively evaluating the performance of both the VIC and the established HC across multiple tasks, including level-ground walking at different speeds, ramps, and stairs, and c) determining whether the VIC could overcome the primary shortcomings of the HC. To achieve these aims, EMG signals from the antagonist muscle pair and ankle sensor inputs were used to drive the muscle models in both controllers, and biomechanical testing was conducted with a single transtibial amputee using a custom-made prosthetic assembly equipped with an electrode interface for recording muscle activity.

II. Methods

A. Surface Electromyography Signals: Acquisition and Processing

A custom-made prosthetic assembly, which is equipped with an electrode interface for detecting muscle activity, was used to record the surface EMG signals of the TA and GAS muscles. Three electrodes were used for each muscle: two recording electrodes arranged in a bipolar configuration and one ground/reference electrode positioned slightly offset. The electrode interface includes a dome-shaped electrode that is connected to the prosthetic liner, along with an electrical conduit that transmits signals from the user's skin surface to the outside of the liner, where it connects to the prosthetic socket [46]. EMG signals were prefiltered in the amplifier using a 20 Hz high-pass filter and a 50 Hz notch filter. The signals were rectified and smoothed to create a linear envelope with a 2 Hz low-pass Butterworth filter. The signals were recorded at a frequency of 1000 Hz and normalized to the user's maximal voluntary contraction (MVC) to generate a control signal. Additionally, an adaptive moving average filter, designed by Sverrisson and Sigurðardóttir [46], was applied to remove artifacts. The time-varying nature of the EMG signal, especially in dynamic conditions, makes relying solely on the normalized activation signal insufficient [47]. Therefore, the activation dynamic model described in [47] was implemented into the system.

B. Finite-State Machine Impedance Controller

Controlled plantar flexion (CP), controlled dorsiflexion (CD), powered pushing (PP), and swing (SW) are the four states of the FSM used in this research. In each, the torque is expressed in (1):

$$\tau_{FSM} = k_n\left(\theta_a - \theta_n^*\right) + b_n\dot{\theta}_a \tag{1}$$

where the ankle angle and angular velocity are represented by θ_a and $\dot{\theta}_a$, while k_n, b_n, and θ_n^* represent the stiffness coefficient, damping coefficient, and equilibrium point, respectively, of the impedance controller of each state [36], [37]. For safety, the PP state is not included in activities involving the descent of stairs and ramps [9], [44].

C. Neuromuscular Model

In the NM employed in [44], muscle contraction dynamics and MSK geometry form the foundation. The muscle-tendon unit (MTU) contraction dynamics were simulated using the Hill-type model, while MSK geometry incorporated the anatomical structure, including joint angles, moment arms, and muscle attachments, to accurately represent force transmission and ankle movements. Together, these components enabled the NM model to simulate realistic muscle behavior and movements [44], [48], [49], [50], [51].

The MTU consists of two primary components: an active contractile element (CE) representing the muscle fiber and a series elasticity (SE) representing the tendon [48]. When the CE is stretched beyond its optimum length, parallel elasticity (PE) is engaged, whereas buffer elasticity (BE) prevents collapse when the SE is equal to slack length. The MTU force F_{MTU} is given by:

$$F_{MTU} = F_{SE} = F_{CE} + F_{PE} - F_{BE} \tag{2}$$

where F_{SE}, F_{CE}, F_{PE}, and F_{BE} are the forces produced by the SE, CE, PE, and BE, respectively, and F_{MTU} is the total isometric muscle force [44]. The force generated by SE is modeled as [50]:

$$F_{SE}\left(\lambda_{SE}\right) = F_{max}\left(\left(\lambda_{SE} - 1\right)/\varepsilon_{ref}\right)^2, \left(\lambda_{SE} > 1\right) \tag{3}$$

where F_{max} is the maximum isometric force that the muscle can produce ε_{ref} is the reference strain and $\lambda_{SE} = l_{SE}/l_{slack}$ is the tendon length normalized to its slack length in (3). The piecewise exponential-linear tendon stiffness curve that is commonly described was approximated using this quadratic form, which aimed to reduce the overall number of model parameters [48]. The unidirectional force generated by the CE is defined as [51]:

$$F_{CE} = F_{max} a f_l\left(l_{CE}\right) f_v\left(v_{CE}\right) \tag{4}$$

where $f_l\left(l_{CE}\right)$ is the normalized fiber force based on the length of CE, l_{CE}, $f_v\left(v_{CE}\right)$ is the normalized velocity fiber force based on the CE's contractile velocity, and a is the muscle activation level. Additionally, the force-length relationship for the high-limit parallel elasticity PE, which is parallel to the CE, is provided below [51]:

$$F_{PE}\left(\lambda_{CE}\right) = F_{max}\left(\left(\lambda_{CE} - 1\right)/w\right)^2, \left(\lambda_{CE} > 1\right) \tag{5}$$

where w represents the width of the bell-shaped curve and $\sigma_{CE} = v_{CE}/v_{max}$ is the contractile element's velocity normalized to the maximal contraction velocity v_{max}. Based on [51], the form of the nonlinear spring is given by a low-limit buffer elasticity (BE):

$$F_{BE}\left(\lambda_{CE}\right) = F_{max}\left(\left(\lambda_{CE} - 1 + w\right)/\left(w/2\right)\right)^2, \left(\lambda_{CE} \le 1 - w\right) \tag{6}$$

The torque generated by the muscle $\tau_{m-muscle}$ is provided below (7), where $R\left(\theta_a\right)$ is a variable moment arm resulting from the muscle attachment to the ankle joint model and θ_a is the ankle angle.

$$\tau_{m-muscle} = F_{MTU} R\left(\theta_a\right) \tag{7}$$

To ensure the model accurately predicts muscle behavior, a surrogate-based optimization method was used to optimize the model parameters [52], and the cost function [53] was defined as the squared error between the biological ankle torque and the model-predicted ankle torque. Optimization was performed using the dataset introduced by Camargo et al. [54], which contains kinematic, kinetic, and EMG measurements from 22 non-disabled adults across a variety of locomotion tasks (level ground, treadmill walking, stair and ramp ascent/descent).

In the joint model, the MTU stiffness K_{mt} is represented as the series combination of the muscle fiber stiffness K_m and the tendon stiffness K_t [38], [40], [41]:

$$K_{mt} = \frac{K_m K_t}{K_m + K_t} \tag{8}$$

TABLE I
FSM IMPEDANCE CONTROL PARAMETERS

Subphase	$k_n\,(Nm/deg)$	$b_n\,(Nms/deg)$	$\theta_n^*\,(deg)$
CP	1	0.02	0
CD	1.5	0.005	0
PP	5	0	-10
SW	2	0	0

K_t is calculated as the local slope of the force–strain curve of the series elastic element (SE), based on the tendon force model provided in (3). According to Cui et al. [55], K_m is determined by the force generated by the muscle F_m, its fiber length at maximum activation l_{opt}, and a scaling constant γ set to 23.4.

$$K_m = \frac{\gamma F_m}{l_{opt}} \tag{9}$$

The active stiffness K_{est} provided by each muscle, considering dynamic variations in muscle-tendon unit length and joint angles, is estimated as [38] and [40]:

$$K_{est} = K_{mt} R^2(\theta_a) + \frac{\partial R(\theta_a)}{\partial \theta_a} F_{MTU} \tag{10}$$

where F_{MTU} is the total force transmitted by the MTU, as defined in (2), and $R(\theta_a)$ is the ankle-angle-dependent moment arm function used to compute joint torque in (7).

D. Hybrid Controller

During the CD and PP subphases, the total torque (τ_t) (11) for the HC was equal to the torque of the NM controller (τ_m) in addition to the FSM impedance controller torque (τ_{FSM}).

$$\begin{aligned}&\tau_t\left(t, \theta_a(t), a(t), \dot{\theta}_a(t)\right)\\ &= \tau_m(t, \theta_a(t), a(t)) + \tau_{FSM}(t, \theta_a(t), \dot{\theta}_a(t))\end{aligned} \tag{11}$$

The NM torque $\tau_m(t)$ was calculated by subtracting the torque generated by the TA muscle τ_{m-TA} from that of the GAS muscle τ_{m-GAS}. During the PP subphase, the NM torque was modeled solely by the negative torque of the GAS muscle [44], [56]. During CP, the controller deactivated the NM model and relied entirely on the impedance controller to prevent excessive plantar flexion [45]. The controller's behavior during the SW phase was represented as position control of the ankle joint around the state, as defined in [57]. The impedance control parameters for the subphases of the FSM controller are listed in Table I.

These FSM impedance control parameters were first tuned separately, before the addition of the $\tau_m(t)$—to optimize user comfort and stability during walking. The tuning process involved adjusting the stiffness and damping values for each subphase based on participant feedback and direct observation. The NM controller software was developed using the MATLAB Coder extension by exporting the optimized MATLAB-Simulink NM model controller as C++ library for controller implementation [45].

TABLE II
OPTIMIZED PARAMETERS IN STIFFNESS TREND FUNCTIONS FOR GAS AND TA MUSCLES

Muscle	b_1	b_2	b_3	b_4	b_5
TA	0.6415	5.9014	0.2512	-21.7250	0.0447
GAS	0.7224	8.1496	0.2778	21.0549	-0.0460

E. Variable Impedance Controller

The VIC was designed to adjust the k_n in the impedance control law in (1) (Section II-B) for CD and PP subphases over time, in response to EMG signals of the residual muscles and the ankle angle.

The mapping between the total muscle activation level and the joint stiffness trend $c_h(t)$ is described by a hyperbolic tangent-like function [42], [43]:

$$c_h(t) = b_1 \frac{1 - e^{-b_2(a(t))}}{1 + e^{-b_2(a(t))}} \tag{12}$$

in which b_1 represents the scaling factor for the influence of muscle activation on $c_h(t)$, b_2 defines the steepness of the response curve, and $a(t)$ is the muscle activation level.

Although this mapping gives a smooth and continuous correlation between muscle activation and joint stiffness, it does not consider the ankle angle, which is necessary for accurately determining muscle stiffness as described in NM (Section II-C). To address this, the following logistic sigmoid function for $c_h(t)$ was introduced, which offers a continuous, bounded, and easily tunable mechanism for transitioning stiffness between minimum and maximum values [58]:

$$c_h(t) = b_1 \frac{1 - e^{-b_2(a(t))}}{1 + e^{-b_2(a(t))}} b_3 \frac{1}{1 + e^{-b_4(\theta_a - b_5)}} \tag{13}$$

where b_3 represents the scaling factor for the influence of the ankle angle, b_4 determines the steepness of the response to changes in ankle angle, and b_5 sets the midpoint of the ankle angle's effect on $c_h(t)$. Although b_1 and b_3 both act as scaling factors and could be combined into a single parameter to avoid redundancy, they were kept separate in our formulation to allow for independent tuning and interpretation of the effects of muscle activation and ankle angle on joint stiffness. This formulation estimates $c_h(t)$ integrating the muscle activation and ankle angle provides a more comprehensive model.

The optimal parameters for $c_{h-TA}(t)$ and $c_{h-GAS}(t)$ functions, representing the TA and GAS stiffness trends, were determined using MATLAB's nonlinear regression analysis function, part of the Statistics and Machine Learning Toolbox [59]. These functions were fitted to the normalized stiffness output of the optimized NM model described in Section II-C. The NM model used the dataset provided by Camaro et al. [54] as its input data. A constraint to the selection of the parameters is that $0 < c_h(t) \leq 1$. The accuracy of the c_h for the TA and GAS muscles prediction models was evaluated using the coefficient of determination (R^2) across ten different trials. On average, the model achieved an R^2 of 97.36% for the GAS muscle and 94.34% for the TA muscle. The optimized parameters for c_{h-TA} and c_{h-GAS} are listed in Table II.

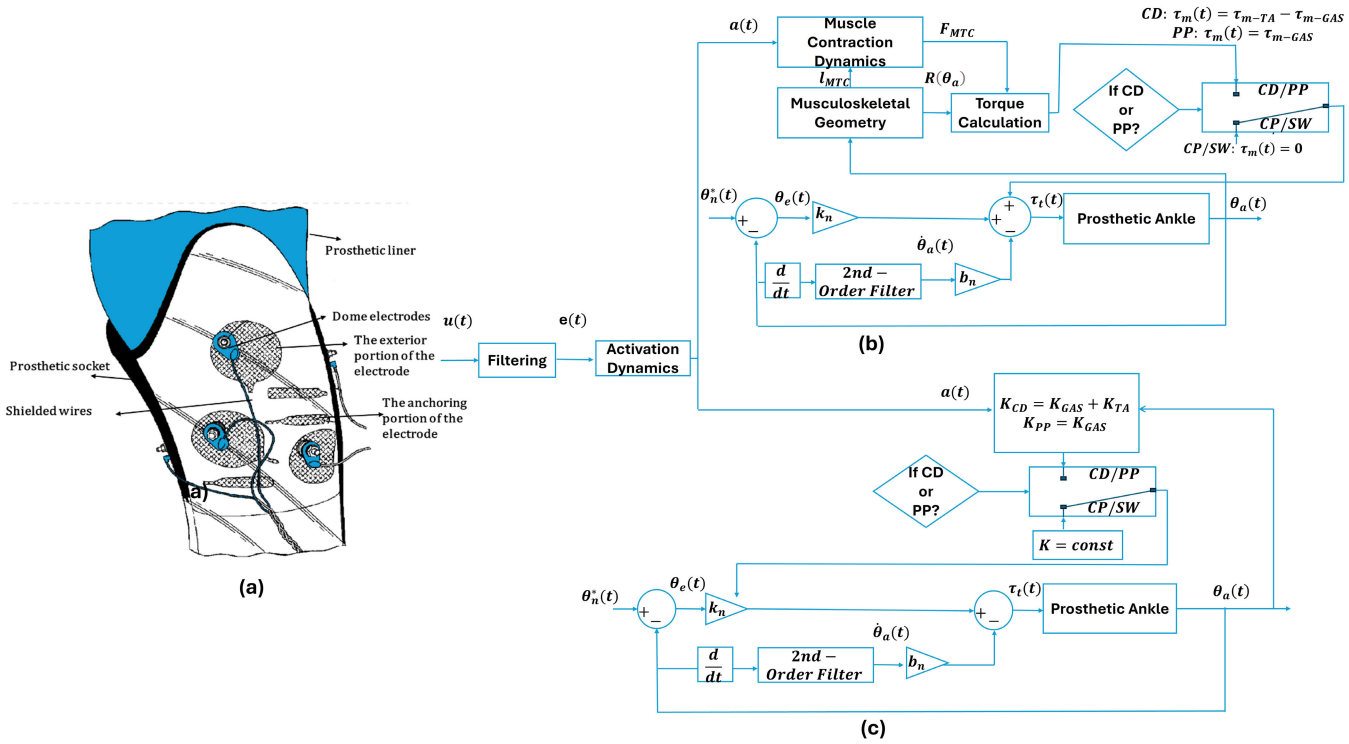

Fig. 1. Overview of the system's components and control systems. The EMG signals from (a) prosthetic assembly having an electrode interface for recording muscle activity [46] were used in both controllers. These signals passed through activation dynamics, where $u(t)$ contains the EMG signals from both muscles, $e(t)$ the filtered EMG signals, and $a(t)$ the activation levels. The EMG control was activated during the CP and PP subphases for both (b) the hybrid controller and (c) the variable impedance controller. In these controllers, the ankle angle and angular velocity are represented by $\theta_a(t)$ and $\dot{\theta}_a(t)$, while k_n, b_n, and θ_n^* represent the linear stiffness, damping coefficient, and equilibrium angle, respectively, of each subphase.

TABLE III
MAXIMUM AND MINIMUM STIFFNESS VALUES (k_{max-n} AND k_{min-n}) FOR EACH SUBPHASE

Subphase	k_{max-n} (Nm/deg)	k_{min-n} (Nm/deg)
CD	3	1
PP	6	3.5

The stiffness of each muscle k_{muscle} for a given subphase was then computed as shown in (14), where k_{max-n} and k_{min-n} refer to the maximum and minimum values for stiffness during each subphase.

$$k_{muscle} = c_{\mathrm{h}}(t)\,(k_{max-n} - k_{min-n}) + k_{min-n} \tag{14}$$

The stiffness values for the CP and SW were kept fixed to align with the HC implementation for comparison. The stiffness value for the CD (k_{CD}) was calculated by adding contributions from both the GAS and TA muscles ($k_n = k_{CD} = k_{GAS} + k_{TA}$). During the PP, the stiffness k_{PP} was derived solely from the GAS muscle ($k_n = k_{PP} = k_{GAS}$). The corresponding for k_{max-n} and k_{min-n} values for CD and PP subphase are listed in Table III. The stiffness values for the CP and SW subphases are consistent with those used in the HC.

In summary, as explained in Section II-D, the HC calculates joint torque as the sum of the FSM impedance torque and NM torque, with impedance parameters, including stiffness (k_n), set as fixed constants for each FSM subphase. In contrast, the VIC also uses the FSM impedance control law but continuously adjusts the stiffness parameter k_n in real time, based on EMG signals and ankle angle, using the mapping introduced in Section II-E. The system components and block diagrams of both HC and VIC controllers are illustrated in Fig. 1.

F. Experimental Protocol

The Icelandic National Bioethics Committee approved the protocol for user testing CII2019061252. The study included a participant with unilateral transtibial amputation who could walk without aids, comprehended all instructions, and had no known neurological or musculoskeletal disorders that could impact the study. The participant was a male, 185 cm tall and weighing 78 kg. He has been living with an amputation for 21 years, which resulted from trauma. The participant gave written informed consent to participate in the research.

Four tasks were employed to compare the performance of the HC and VIC: ramp ascent (RA) and ramp descent (RD), followed by level-ground walking (LGW) at self-selected walking speed (SSWS) and high walking speed (HWS), and finally stair ascent (SA) and stair descent (SD). Approximately twenty strides were recorded for each controller and task (RA, RD, LGW at SSWS, and HWS), and six trials were performed for each controller for stair ascent and descent. Stair tasks were performed on a three-step staircase with a step height of 155 mm and an incline of 35 deg, while ramp tasks were performed on a 7.5 deg incline. For each activity, all trials

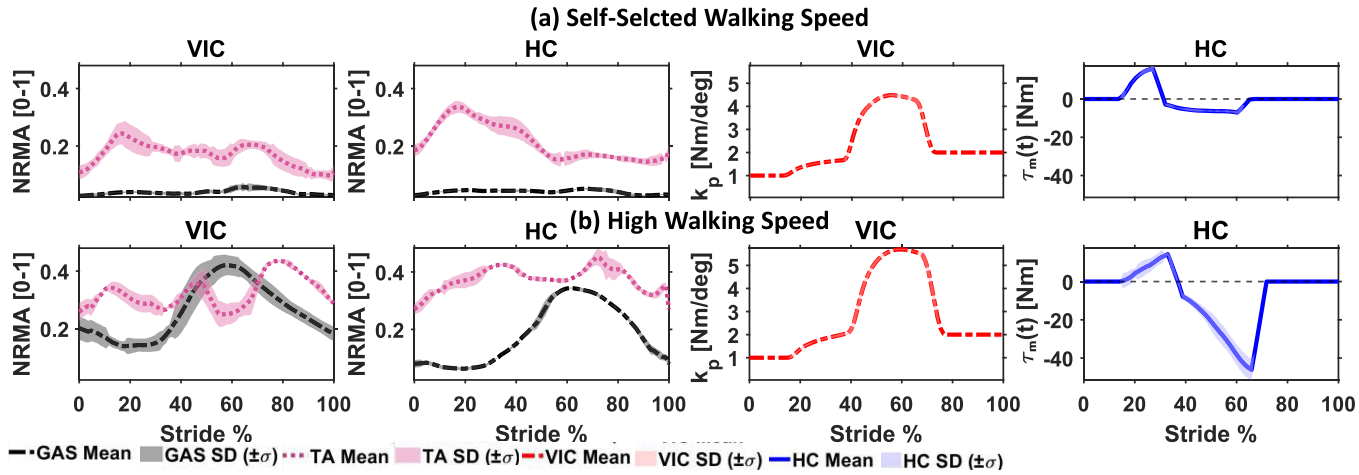

Fig. 2. NRMA data for both controllers, alongside stiffness value gain $\boldsymbol{k_p(t)}$ for the VIC and NM torque $\boldsymbol{\tau_m(t)}$ for the HC motor inputs, averaged over strides ±1 standard deviation (shading) during LGW for (a) SSWS and (b) HWS conditions.

were first performed using one controller and then repeated with the other, with the sequence of controllers alternated each time. The participant was informed which controller was in use for each set of trials, as a safety measure. Rest periods were provided between activities as needed, with the total session lasting approximately three hours.

For the comparison, eight different parameters were analyzed: ankle angle $\theta_a(t)$, ankle torque $\tau_a(t)$, ground reaction forces (GRF) in the vertical direction, ankle power, normalized residual muscle activity (NRMA) for both controllers, stiffness value gain $k_p(t)$ for the variable impedance controller, and the NM torque $\tau_m(t)$ for the HC. Additionally, user feedback on perceived performance and control experience was informally collected following the session to support interpretation of the biomechanical results.

Data collection utilized a 3D motion capture system (Vicon, Oxford, UK) with ten cameras and six force plates (AMTI, MA, USA), sampling at 100 Hz and 1000 Hz, respectively. A customized lower-body marker set was employed to create a six-degree-of-freedom model. Data was processed using VICON Nexus and Cleanse (Moveck Inc.) software. In this research, a powered ankle prosthesis with a maximum motor output of $75Nm$ was used and designed to operate in parallel with carbon fiber blades built on the Össur Pro-Flex Pivot (category 5). The parallel configuration was achieved via a custom double-link mechanism. The participant was already familiar with using the same powered ankle prosthesis and had experience performing a variety of activities, including level walking, navigating ramps, and climbing stairs, with this system. Before data collection, a dedicated two-hour preparation and training period was provided to practice all activities further using both controllers.

III. Results

A. Level-Ground Walking

Biomechanical and EMG data were collected during the LGW for both controllers (VIC and HC) under two speed conditions: SSWS and HWS. Fig. 2 presents NRMA data for both controllers, along with $k_p(t)$ for the VIC and $\tau_m(t)$ for the HC motor inputs during LGW at SSWS (Fig. 2a) and HWS (Fig. 2b). At SSWS, the mean speed for the VIC was 1.34 m/s compared to 1.35 m/s for the HC, with corresponding cadences of 113.7 and 115.6 steps/min, respectively. At HWS, the VIC and HC achieved mean speeds of 1.82 m/s and 1.91 m/s, respectively, and cadences of 132.2 and 136.1 steps/min.

During SSWS walking trials the GAS muscle showed minimal activation with both controllers, with peak NRMA values below 0.1 (Fig. 2a). During HWS walking, the user showed greater GAS muscle engagement, as indicated by an increase in NRMA to a peak value 0.42 ± 0.024 with the VIC and 0.33 ± 0.018 with the HC (Fig. 2b). The VIC demonstrated a higher k_p value, reaching $5.69 \pm 0.012 Nm/deg$, compared to $4.48 \pm 0.053 Nm/deg$ for the HC (Fig. 2). The NM torque $\tau_m(t)$ of the HC reached a much higher value at HWS ($-45.33 \pm 2.48 Nm$) than SSWS ($-7.02 \pm 0.64 Nm$). Unless otherwise specified, values are reported as mean ± standard deviation (SD) across strides.

Fig. 3 illustrates a comparison of biomechanical data, including $\theta_a(t)$ along with the kinematic data averaged from 10 non-amputee subjects for SWSS and HWS, as reported by [60], $\tau_a(t)$, GRF, and power for both controllers during LGW at SSWS and HWS. Table IV presents a comprehensive comparison of ankle biomechanical parameters between the two controllers, including range of motion (ROM), peak ankle torque (τ_a), peak power, and the first and second peaks of ground reaction force (GRF) LGW at SSWS and HWS. To further quantify the magnitude of differences between the controllers, Cohen's d effect sizes were calculated for each metric, providing a standardized measure of the effect independent of sample size [61].

B. Ramp Ascent and Descent

Biomechanical and EMG data were collected during RA and RD at a 7.5 deg incline for both controllers (VIC and HC). NRMA data for both controllers $k_p(t)$ for the VIC and $\tau_m(t)$ for the HC motor inputs during RA and RD are shown in Fig. 4. For RA, mean speeds were 1.20 m/s (VIC) and 1.39 m/s (HC), with cadences of 102.9 and 107.5 steps/min.

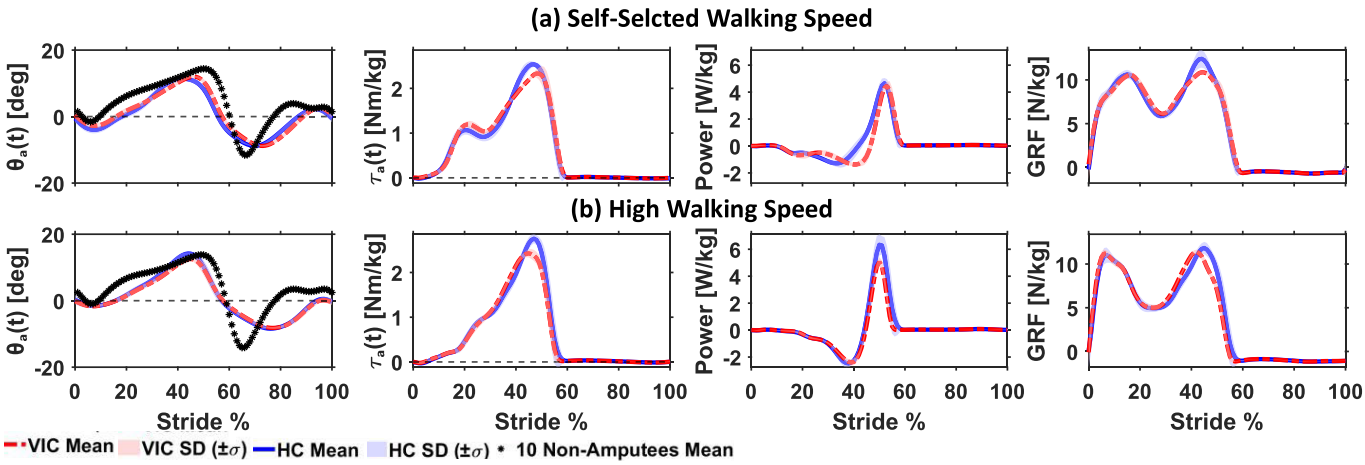

Fig. 3. Comparison of biomechanical results, including ankle angle $\theta_a(t)$, ankle torque $\tau_a(t)$, GRF, and power for both controllers, averaged over strides ±1 standard deviation (shading) during LGW testing for (a) SSWS and (b) HWS conditions.

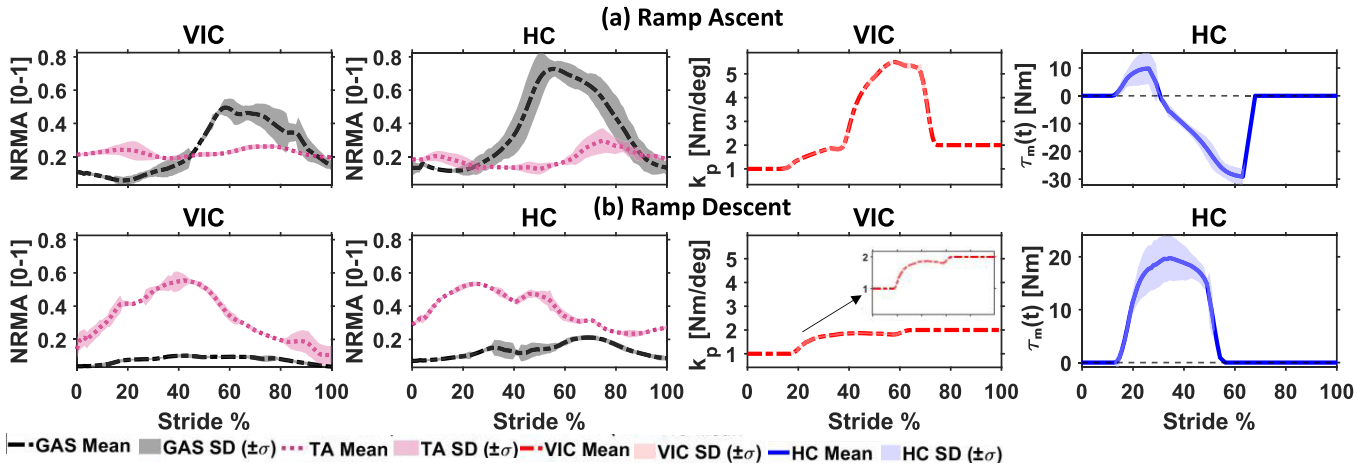

Fig. 4. NRMA data for both controllers, alongside *stiffness value gain* $k_p(t)$ *for the VIC and NM torque* $\tau_m(t)$ for the HC motor inputs, averaged over strides ±1 standard deviation (shading) during (a) RA and (b) RD.

TABLE IV
COMPARISON OF ANKLE BIOMECHANICAL PARAMETERS BETWEEN VIC AND HC CONTROLLERS DURING LGW AT VARIOUS SPEEDS

Controller	ROM (deg)	Peak τ_a (Nm/kg)	Peak *Power* (W/kg)	GRF 1st peak (N/kg)	GRF 2nd peak (N/kg)
SSWS					
VIC	20.74 ± 1.41	2.33 ± 0.08	4.46 ±0.31	10.57 ±0.26	10.87 ±0.22
HC	20.32 ± 1.50	2.52 ± 0.091	4.65 ±0.40	10.62 ±0.52	12.21 ±0.98
Cohen's d	0.29	−2.21	−0.53	−0.12	−1.86
HWS					
VIC	20.82 ± 1.68	2.43 ± 0.10	4.73 ±0.45	11.16 ±0.41	11.34 ±0.26
HC	22.61 ± 1.83	2.75 ± 0.11	6.63 ±0.78	11.02 ±0.87	11.83 ±0.70
Cohen's d	−1.02	−3.05	−2.92	0.21	−0.96

For RD, mean speeds were 1.23 m/s (VIC) and 1.28 m/s (HC), with cadences of 117.2 and 114.8 steps/min, respectively.

During RA (Fig. 4a), the NRMA of the GAS muscle was notably high for both controllers, reaching peak values 0.49 ± 0.053 for the VIC and 0.72 ± 0.056 for the HC. This increased muscle activation influenced the proportional gain k_p value, which was $5.50 \pm 0.057 Nm/deg$ for the VIC. For the HC, the maximum value of $\tau_m(t)$ during plantar flexion was $-29.10 \pm 2.88 Nm$. Both values are below the values observed during LGW at HWS. In contrast, during the RD (Fig. 4b), the TA muscle was higher, reaching peak values for NRMA of 0.55 ± 0.039 for the VIC and 0.53 ± 0.027 for the HC.

The k_p value increased gradually during the CD, transitioning from CP to SW values, ranging from $1 to\ 2 Nm/deg$. There was no PP subphase, as described in Section II-B. For the HC, the $\tau_m(t)$ reached peak dorsiflexion torque of $-19.70 \pm 4.22 Nm$, with a relatively high standard deviation. Fig. 5 shows a comparison of biomechanical data for both controllers during RA (Fig. 5a) and RD (Fig. 5b) along with the kinematic data averaged from 10 non-amputee subjects for RA and RD at a 5 deg incline, as reported by [60]. The two controllers' biomechanical data for the RA and RD are compared in Table V, including ROM, peak τ_a, peak power, and the first and second peaks of GRF.

C. Stair Ascent and Descent

Biomechanical and EMG data were collected during SA and SD for both controllers (VIC and HC). Fig. 6 presents

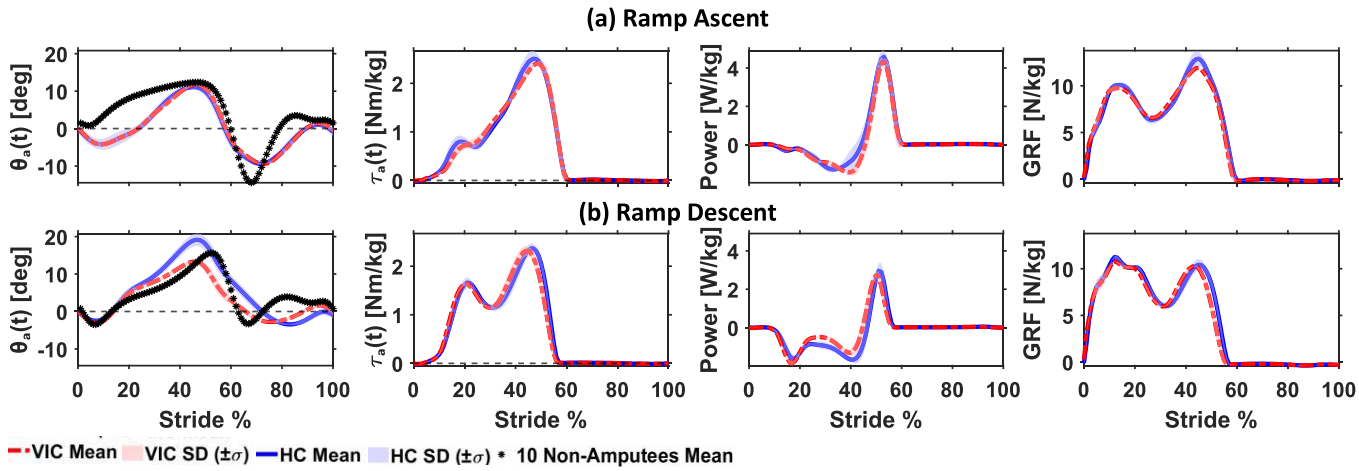

Fig. 5. Comparison of biomechanical results, including ankle angle $\boldsymbol{\theta_a(t)}$, ankle torque $\boldsymbol{\tau_a(t)}$, GRF, and power for both controllers, averaged over strides ±1 standard deviation (shading) during (a) RA and (b) RD.

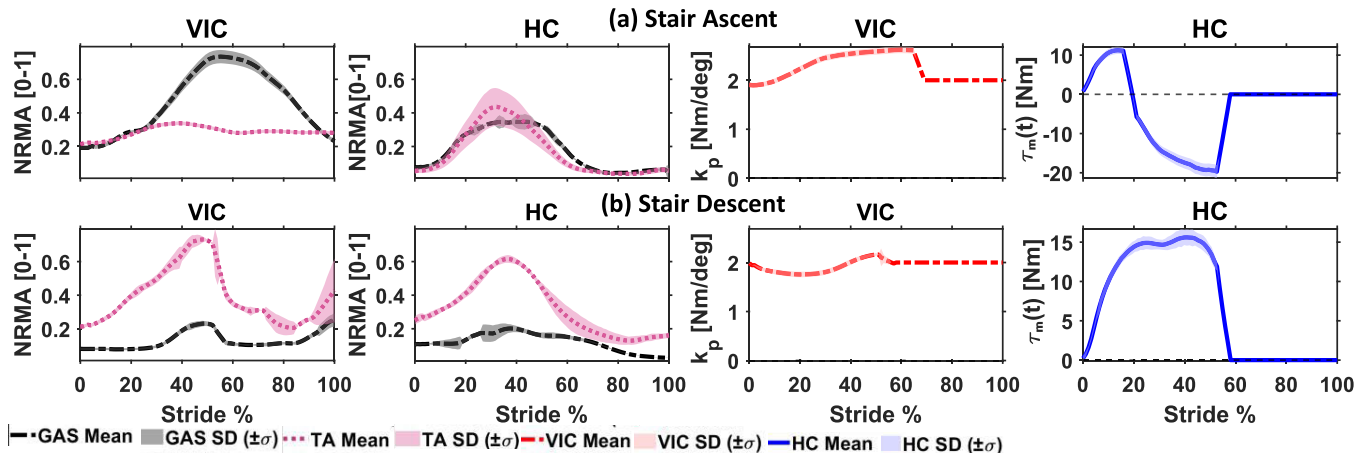

Fig. 6. NRMA data for both controllers, stiffness value gain $\boldsymbol{k_p(t)}$ for the VIC and NM torque $\boldsymbol{\tau_m(t)}$, averaged over strides ±1 standard deviation (shading) during (a) SA and (b) SD.

TABLE V
COMPARISON OF ANKLE BIOMECHANICAL PARAMETERS BETWEEN VIC AND HC CONTROLLERS DURING RA AND RD

Controller	ROM (deg)	Peak τ_a (Nm/kg)	Peak $Power$ (W/kg)	GRF 1st peak (N/kg)	GRF 2nd peak (N/kg)
RA					
VIC	21.11 ± 1.76	2.41 ± 0.06	4.32 ±0.14	9.77 ±0.21	11.90 ±0.06
HC	20.51 ± 2.17	2.51 ± 0.18	4.58 ±0.32	10.09 ±0.23	12.71 ±0.68
Cohen's d	0.30	−0.77	−1.04	−1.45	−1.69
RD					
VIC	16.33 ± 1.89	2.32 ± 0.13	2.74 ±0.18	10.81 ±0.02	10.28 ±0.04
HC	22.60 ± 2.14	2.36 ± 0.10	2.95 ±0.42	11.19 ±0.20	10.43 ±0.51
Cohen's d	−3.10	−0.33	−0.66	−2.71	−0.42

the NRMA data for both controllers, $k_p(t)$ for the VIC and $\tau_m(t)$ for the HC motor input during SA and SD. For SA, the mean cadence was 98 and 107 steps/min, with corresponding horizontal speeds of 0.36 and 0.39 m/s. For SD, the mean cadence was 105 and 111 steps/min, with horizontal speeds of 0.39 and 0.41 m/s, respectively. During SA (Fig. 6a), the GAS muscle activation level for the VIC was notably high (0.7351 ± 0.04), compared to lower activation levels with the HC (0.35 ± 0.03). The activity of the TA muscle was slightly higher with the HC. Furthermore, the user did not load the heel at initial contact, so the FSM transitioned directly into CD, leading to a gradual increase in the k_p value for the VIC starting from the SW phase, reaching a maximum of $2.62 \pm 0.07 Nm/deg$. The maximum k_p value was greatly affected by the controller's inability to transition to the PP subphase. For the HC, the maximum value of $\tau_m(t)$ for the HC during dorsiflexion was $11.20 \pm 0.94 Nm$ and a maximum plantar flexion value of $-19.55 \pm 1.66 Nm$.

During SD (Fig. 6b), the TA muscle activity was noticeably high for both controllers, like the pattern seen during RD. Similar to SA, the user did not load the heel at first contact. The k_p value for the VIC initially decreased before gradually increasing to a maximum of $2.16 \pm 0.07 Nm/deg$. For the HC controller, the $\tau_m(t)$ reached a maximum dorsiflexion torque value of $15.60 \pm 0.98 Nm$.

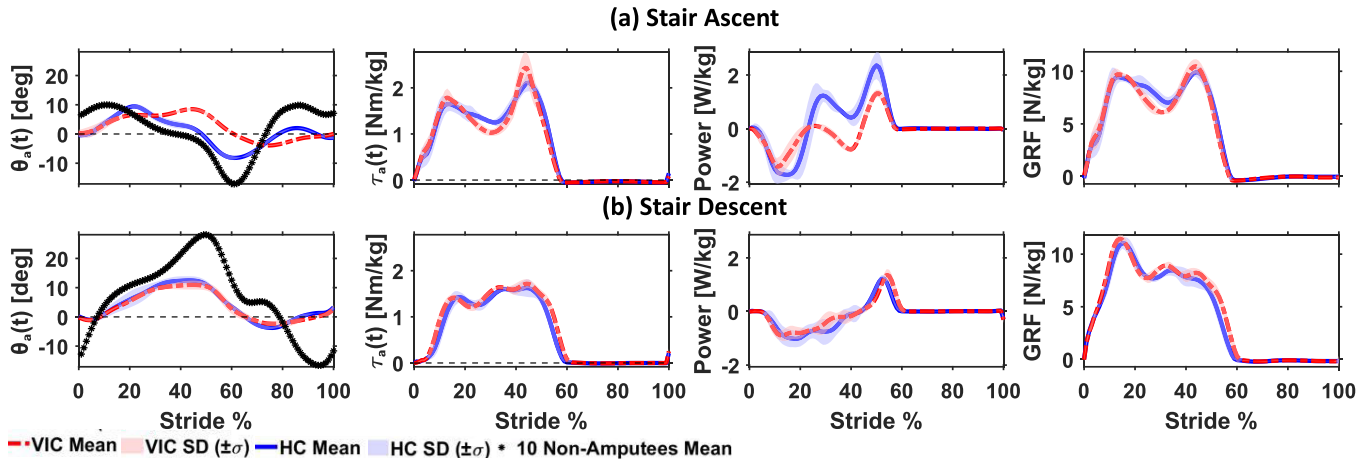

Fig. 7. Comparison of biomechanical results, including ankle angle $\boldsymbol{\theta_a(t)}$,ankle torque $\boldsymbol{\tau_a(t)}$, GRF, and power for both controllers, averaged over strides ±1 standard deviation (shading) during (a) SA and (b) SD.

TABLE VI
COMPARISON OF ANKLE BIOMECHANICAL PARAMETERS BETWEEN VIC AND HC CONTROLLERS DURING SA AND SD

Controller	ROM (deg)	Peak τ_a (Nm/kg)	Peak $Power$ (W/kg)	GRF 1st peak (N/kg)	GRF 2nd peak (N/kg)
SA					
VIC	12.43 ± 1.86	2.44 ± 0.34	1.34 ±0.14	9.69 ±0.54	10.46 ±0.69
HC	18.63 ± 1.53	2.12 ± 0.18	2.42 ±0.51	9.44 ±0.65	9.91 ±0.45
Cohen's d	−3.65	1.19	−2.92	0.42	0.95
SD					
VIC	13.31 ± 1.86	1.73 ± 0.097	1.39 ±0.22	11.38 ±0.42	8.92 ±0.25
HC	16.26 ± 2.83	1.65 ± 0.19	1.22 ±0.14	11.03 ±0.53	8.46 ±0.50
Cohen's d	−1.23	0.53	0.89	0.73	1.12

A comparison of the biomechanical data for both controllers during SA and SD, along with kinematic data averaged from 10 non-amputee subjects as reported by [60], is presented in Fig. 7. Table VI compares the biomechanical data from the two controllers for the SA and SD.

IV. DISCUSSION

A. Level-Ground Walking

During LGW at HWS, the user was contracting his GAS muscle more compared to SSWS, which, coupled with ankle movement, affected how both controllers responded to the dynamic demands. For the VIC, the k_p value reached a higher peak, and for the HC, the τ_m reached a higher peak plantar flexion torque value.

Biomechanical results showed that both controllers produced higher power and torque outputs at HWS compared to SSWS, consistent with previous research [62]. However, the HC consistently produced considerably greater peak torque and power than the VIC, even though the user achieved close identical walking speeds (1.82 m/s VIC, 1.91 m/s HC) and cadences (132.2 vs. 136.1 steps/min), as reported in Section III-A. This is especially notable at HWS, where the peak torque (d = −4.05) and peak power (d = −2.92) Cohen's d effect sizes were extremely large, highlighting a substantial difference between controllers, as reported in Table IV. For these metrics, the HC also exhibited a higher standard deviation, indicating its greater performance variability. In contrast, the VIC demonstrated greater repeatability, likely due to employing a single equation for the stiffness calculation, whereas the HC's complexity with multiple equations and interdependent parameters made it more sensitive to minor changes or uncertainties in system parameters. These findings were supported by user feedback, which noted that while the HC felt more responsive to muscle contractions, it lacked consistency in prosthetic control compared to the VIC.

Force plate data showed that vertical GRF consistently increased with speed for the VIC, whereas the HC exhibited higher standard deviation and a noticeably higher peak GRF at SSWS. This suggests that the pronounced GRF peak with the HC was due to the user exerting greater force to maintain stability in response to the controller's variability.

The HC helped the user achieve more dorsiflexion during the CD subphase at HWS compared to the VIC. At SSWS, however, the maximum dorsiflexion angle was reached faster, due to the dorsiflexion $\tau_m(t)$ torque provided by the HC during the CD subphase. Data from non-amputee subjects [60] indicated a delayed maximum dorsiflexion, with the VIC more closely aligning with this pattern. Notably, both controllers showed a reduced plantar flexion angle in the CP subphase during LGW at higher speeds, consistent with published research [62]. Additionally, non-amputee kinematic data showed a higher maximum plantar flexion angle than both controllers, as both controllers are designed to reach a maximum of 10 degrees during the PP subphase (Fig. 3B).

B. Ramp Ascent and Descent

During RA, muscle activation was higher than that found during LGW. Both k_p and τ_m reached higher peak values than during LGW at SSWS, but lower than LGW at HWS, despite greater levels of muscle activation, highlighting the key role of ankle angle in both controllers. The power output of the

HC was substantially higher than that of the VIC, as indicated by the large effect size for peak power (Cohen's $d = -1.04$), while torque was only moderately higher and showed greater variability, similar to LGW, as reported in Table V. The second peak of the vertical GRF was notably high with a high standard deviation, as was found during LGW, reflecting the user's adaptation to the inconsistencies of the HC.

During RD, both controllers demonstrated reduced power and torque compared to RA, which was expected and consistent with the literature in [48], where stability and controlled movement are prioritized over propulsion. The ROM was significantly higher in the HC than with the VIC, primarily due to higher dorsiflexion angles. The dorsiflexion torque input from the NM was the primary cause of this increased dorsiflexion angle in HC, which resulted in the ankle being less stiff during this phase. In contrast, with the VIC, the ankle became gradually stiffer due to a progressive increase in controller stiffness, thereby limiting dorsiflexion and reducing the overall ROM (Fig. 4b). Anecdotally, the user reported feeling more in control of the VIC during RD despite its reduced ROM, while rapid ankle movement with HC was perceived as destabilizing and mentally demanding. The slightly higher cadence with VIC (117.2 vs. 114.8 steps/min) supports this feedback. This feedback may indicate that the VIC would benefit from increased stiffness to improve user comfort, control, and perceived safety when descending a ramp. Notably, both controllers were aligned with non-amputee kinematic data in showing no pushing during RD (Fig. 5B), due to the disabling of the PP phase during downward activities.

C. Stair Ascent and Descent

During SA, when using the VIC, the user was unable to reach the angle threshold required for transitioning into the PP subphase. This limitation is explained by the high stiffness gain value in the CD phase, which stiffened the ankle and restricted its range of motion. Consequently, the ankle was unable to dorsiflex sufficiently to trigger the PP subphase transition. To compensate for the lack of push-off power assistance, the user increased GAS muscle activation, but since the controller did not transition to the PP phase, the additional muscular effort did not translate into effective propulsion. As a result, the performance of VIC during SA was limited. The HC, on the other hand, performed better by providing the required push-off during SA. By providing more dorsiflexion torque, the HC made it easier for the user to reach the threshold angle and allowed for a smoother transition into the PP subphase. Increased ROM and improved power delivery demonstrated HC's effectiveness in meeting the biomechanical demands of SA. Also, the HC ankle angle trajectory has more closely resembled patterns observed in non-amputee individuals [9], [60].

The results for SD were consistent with those obtained during RD. Despite the HC's higher ROM, the user felt more in control and comfortable with the VIC. However, the kinematic data showed a higher maximum dorsiflexion angle compared to both controllers during ramp descent (Fig. 7B). It is important to note that the direct comparison with non-amputee data serves only as an indication, as the conditions were not the same, including differences in the number of steps, ramp inclination, and step size.

Overall, the participant expressed a preference for the incorporation of EMG signals for more power-demanding activities like RA, SA, and LGW at HWS, compared to less demanding activities. The participant experienced only mild fatigue and no cramps after approximately three hours of testing, which might be attributed to his status as a K-4 level user.

D. Limitations and Future Directions

The VIC was more robust and user-perceived to be controllable, providing consistent and repeatable performance throughout tasks compared to the more unpredictable and variable behavior observed with the HC. However, it was limited in its ability to facilitate smooth transitions between subphases, particularly during SA. In contrast, the HC controller facilitated smoother and more natural transitions between gait subphases by directly adding the neuromuscular (NM) torque to the FSM impedance torque. As a future direction for the VIC, EMG signals could be used to enhance the transition between subphases, potentially improving the controller's responsiveness and alignment with natural movement patterns.

To enable detailed comparison across various tasks and conditions, facilitate iterative controller development, and allow in-depth assessment of user feedback and adaptation throughout the testing process, this study was conducted as a single-subject case study with a high-activity (K4-level) user. To ensure the wider applicability and effectiveness of the VIC, further user testing across a broader range of participants and environments is needed to address the limitations of this initial case study.

V. Conclusion

In this research, a comparison between a novel EMG-driven VIC and an HC that combines an FSM impedance controller with a volitional EMG-driven MSK model was performed. By enabling more robust, adaptive, and user-controllable modulation of ankle stiffness, the VIC was developed to address the primary limitations of the HC, including its excessive and unpredictable torque output and lack of repeatability. Performance was evaluated through biomechanical testing with a transtibial amputee across different activities, such as stair navigation, RA, RD, and LGW. The results showed that the VIC provided better control, lower power output, and more reliable, repeatable performance. This was because of its robust impedance modulation and simplified model structure. Nonetheless, the HC controller performed better in enabling more seamless transitions between gait subphases, particularly during SA.

Conflict of Interest

Anna Lára Ármannsdóttir and Atli Örn Sverrisson are employees of Össur ehf. The funders had no involvement in the study design; data collection, analyses, or interpretation; manuscript preparation, or the decision to publish the findings.

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